D4.1 Natural selection

NAME	DATE	TOTAL
		_____ / 36

About 35 minutes, or one section at a time. The number of marks shows how many separate points to make.

Section A: Recall

6 MARKS

1 STATE the original source of all new alleles. [D4.1.2] [1]

2 LIST two resources that may limit a population. [D4.1.3] [2]

3 STATE two examples of density-independent factors. [D4.1.4] [2]

4 DEFINE fitness. [D4.1.5] [1]

Section B: Understanding

13 MARKS

5 OUTLINE how sexual reproduction creates variation. [D4.1.2] [3]

6 EXPLAIN why overproduction of offspring promotes natural selection. [D4.1.3] [3]

7 EXPLAIN why a giraffe's stretched neck would not be passed to its offspring. [D4.1.6] [2]

8 **EXPLAIN** why males with costly, exaggerated traits may still leave more offspring. [D4.1.7] [3]

9 **EXPLAIN** how an abiotic factor can act as a selection pressure. [D4.1.4] [2]

Section C: Working with data

11 MARKS

All data in this section are invented for practice, using patterns reported in real studies.

10 [D4.1.8]

Guppies bred for 6 months without predators. Then ponds were given no predator or a dangerous predator. The mean number of spots per male was counted.

POND	PREDATORS ADDED	AFTER 5 MONTHS	AFTER 14 MONTHS
NO PREDATORS	10.0	11.5	12.5
DANGEROUS PREDATORS	10.0	9.0	8.5

a **CALCULATE** the percentage change in spots in the pond with no predators, from when predators were added to 14 months. [1]

b **EXPLAIN** the trend where there were no predators. [2]

c **EXPLAIN** the trend with dangerous predators. [2]

d **SUGGEST** one reason why using ponds is better than only observing wild populations. [1]

11 [D4.1.4, D4.1.5]

The percentage of dark moths in one area was recorded as air pollution fell.

YEAR	1960	1975	1990	2005
DARK MOTHS / %	92	70	31	8

a DESCRIBE the trend. [1]

b EXPLAIN the trend using natural selection. [3]

c SUGGEST why a small percentage of dark moths may remain. [1]

Section D: Longer answer

6 MARKS

12 EXPLAIN how natural selection can change the characteristics of a population over many generations. [6]

[D4.1.1–6]